
**Investigating causal
relationship between 4th
down attempts and winning
in the National Football
League (NFL)**

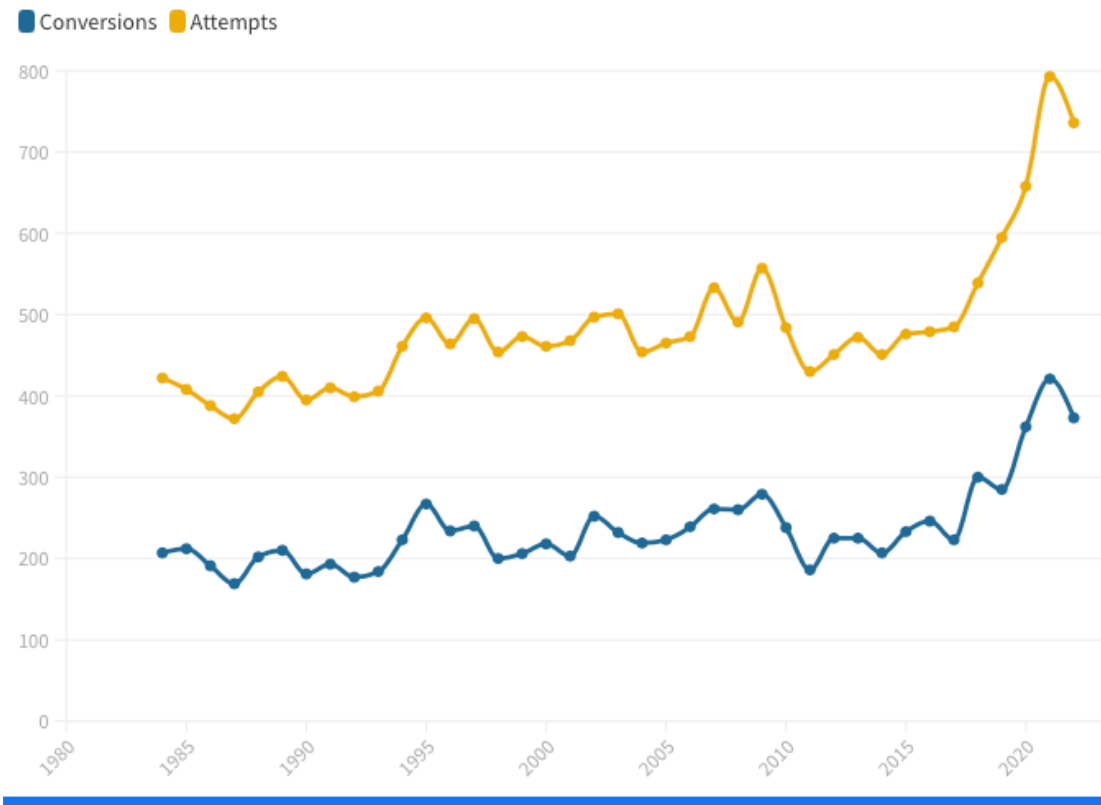
Rules of American Football

- *Objective*: Most amount of points
- *Format*: Offense, Defense, Special Teams
- *Possessions*: Each team has "4 downs" to achieve 10 yards +
- *Plays*: Run, Pass, Kick a Field Goal or Punt
- *4th downs*: Can result in a success, a turnover or kicking the ball away
- *Strategies*: Aggressive (Use all 4 downs), Passive (Use only first 3 downs)

History of 4th down attempts

- "You have to be willing to take risks in this game, but you also have to know when the risks are worth taking. Sometimes, failing on 4th down can cost you more than the game." – Andy Reid
- Belichick 4th and-2 (2009)

NFL Fourth Downs by Year
Conversions and Attempts since 1984



*(Sutelan, 2023)

4th down attempts: Winning and Losing

- *Out of 32 teams NFL teams:*
 - *Top 3 teams in 4th down attempts in 2024:* Browns, Bears, Giants
 - *Bottom 3 teams in 4th down attempts in 2024:* Vikings, Buccaneers, Ravens
 - *Tied most losses in 2024:* Giants and Browns
 - *Top 10 in wins in 2024:* Vikings, Buccaneers and Ravens
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- *Most 4th down attempts in NFL since 2021:* Lions
 - *Top 5 in wins in NFL since 2021:* Lions

Framework

- Used the 2011-2024 NFL regular seasons
- *Analyzed three different teams:* Best (KC), Worst (JAX) and Average (IND) Team since 2011
- *Causal Question:* Does attempting 4th down plays in an NFL game cause more winning?
- *Exposure of Interest:* Running at least one play on 4th down in a given game
- *Outcome:* Winning the game

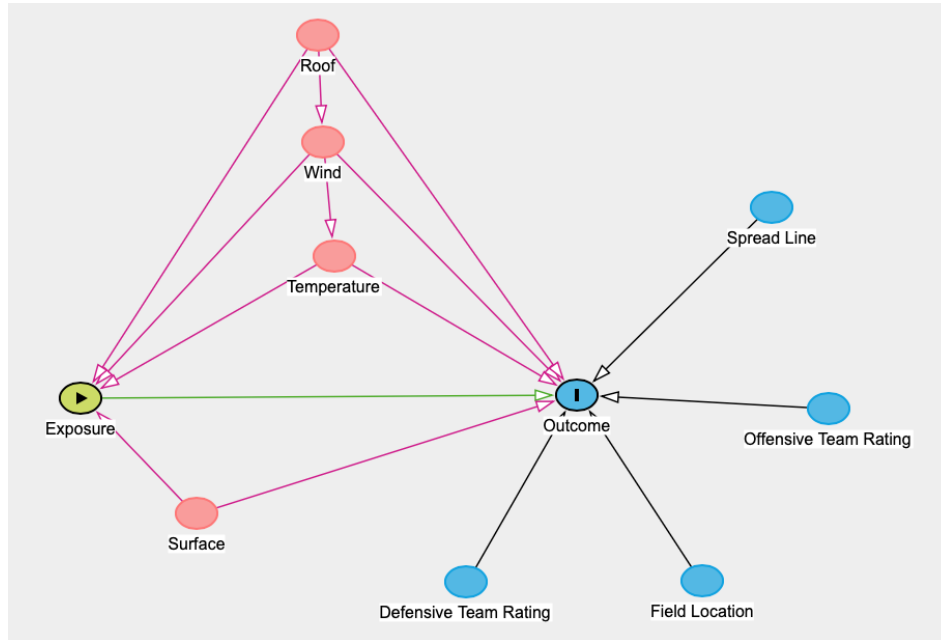
Causal Parameters and Potential Results

- Y^0, Y^1 : Winning an NFL game if there was no 4th down play (punt, field goal) attempted vs if there was a 4th down play (run, pass) attempted (during the 2011-2024 NFL seasons)
- **Marginal Risk Difference (MRD)**: $P(Y^1 = 1) - P(Y^0 = 1)$ = Average difference in the probability of winning if every game had an attempted 4th down play versus if every game had no attempted 4th down play
- **Average Treatment Effect on the Treated (ATT)**: $P(Y^1 = 1 | A = 1) - P(Y^0 = 1 | A = 1)$ = Among games where a team attempted a 4th down play, the average difference in the probability of winning compared to if they had not attempted one
- **Average Treatment Effect on the Control (ATC)**: $P(Y^1 = 1 | A = 0) - P(Y^0 = 1 | A = 0)$ = Among games where a team did not attempt a 4th down play, the average difference in the probability of winning if they had attempted one

Causal Parameters

- *TMLE SL*: ATT, ATC and MRD
- *Matching*: ATT
- *IPTW*: MRD
- Bootstrap for Confidence Intervals

Variables of interest



- Potential Confounders: Wind (mph)*, Temperature (F)*, Roof, Surface
- Variables that impact the outcome: Spread Line*, Opposing Team Defensive Team Rating*, Offensive Team Rating* and Field Location*

* (Yam & Lopez, 2018) "Quantifying the causal effects of conservative fourth down decision making in the National Football League"

Table 1 Indianapolis Colts (IND)

<i>Game characteristics</i>		<i>Exposure = 0 (N = 77)</i>	<i>Exposure = 1 (N = 137)</i>
Temperature (F) ; Moyenne (É.T.)		56,25(16,03)	57,46 (17,72)
Spread line ; Moyenne (É.T.)		-0,39 (6,49)	-0.54 (5,82)
Wind (MPH) ; Moyenne (É.T.)		8,69 (4,19)	7,31 (4,39)
Outcome ; N (%)			
	Loss	26 (33,8%)	82 (59,9%)
	Win	51 (66,2%)	55 (40,1%)
Field Location ; N (%)			
	Away	28 (36,4%)	73 (53,3%)
	Home	49 (63,6%)	64 (46,7%)
Surface ; N (%)			
	Turf	59 (76,6%)	87 (63,5%)
	Grass	18 (23,4%)	50 (36,5%)
Roof ; N (%)			
	Closed/Dome	42 (54,5%)	63 (46,0%)
	Outdoor/Open	35 (45,5%)	74 (54,0%)
Opposing team defensive team rating ; N (%)			
	Top 16	37 (48,1%)	73 (53,3%)
	Bottom 16	40 (51,9%)	64 (46,7%)
Offensive team rating ; N (%)			
	Top 16	38 (49,4%)	100 (73,0%)
	Bottom 16	39 (50,6%)	37 (27,0%)

É.T. = Écart-Type

Table 1 Jacksonville Jaguars (JAX)

<i>Game characteristics</i>		<i>Exposure = 0 (n = 63)</i>	<i>Exposure = 1 (n = 144)</i>
Temperature (F) ; Moyenne (É.T.)		66,27 (15,64)	65,06 (14,51)
Spread line ; Moyenne (É.T.)		-2,44 (5,12)	-3.99 (6,24)
Wind (MPH) ; Moyenne (É.T.)		8,10 (4,06)	8,13 (4,48)
Outcome ; N (%)			
	Loss	29 (46,0%)	113 (78,5%)
	Win	34 (54.0%)	31 (21,5%)
Field Location ; N (%)			
	Away	28 (44,4%)	74 (51,4%)
	Home	35 (55.6%)	70 (48.6%)
Surface ; N (%)			
	Turf	18 (28,6%)	27 (18,8%)
	Grass	45 (71,4%)	117 (81,2%)
Roof ; N (%)			
	Closed/Dome	8 (12,7%)	21 (14,6%)
	Outdoor/Open	55 (87,3%)	123 (85,4%)
Opposing team defensive team rating ; N (%)			
	Top 16	28 (44,4%)	75 (52,1%)
	Bottom 16	35 (55,6%)	69 (47,9%)
Offensive team rating ; N (%)			
	Top 16	20 (31,7%)	39 (27,1%)
	Bottom 16	43 (68,3%)	105 (72,9%)

É.T. = Écart-Type

Table 1 Kansas City Chiefs (KC)

<i>Game characteristics</i>		<i>Exposure = 0 (N = 92)</i>	<i>Exposure = 1 (N = 112)</i>
Temperature (F) ; Moyenne (É.T.)		60,43 (16,89)	53,21 (18,49)
Spread line ; Moyenne (É.T.)		3,85 (5,80)	2,25 (7,37)
Wind (MPH) ; Moyenne (É.T.)		7,86 (4,27)	8,14 (4,95)
Outcome ; N (%)			
	Loss	20 (21,7%)	46 (41,1%)
	Win	72 (78,3%)	66 (58,9%)
Field Location ; N (%)			
	Away	43 (46,7%)	58 (51,8%)
	Home	49 (53,3%)	54 (48,2%)
Surface ; N (%)			
	Turf	23 (25,0%)	25 (22,3%)
	Grass	69 (75,0%)	87 (77,7%)
Roof ; N (%)			
	Closed/Dome	12 (13,0%)	7 (6,2%)
	Outdoor/Open	80 (87,0%)	105 (93,8%)
Opposing team defensive team rating ; N (%)			
	Top 16	44 (47,8%)	46 (41,1%)
	Bottom 16	48 (52,2%)	66 (58,9%)
Offensive team rating ; N (%)			
	Top 16	80 (87,0%)	95 (84,8%)
	Bottom 16	12 (13,0%)	17 (15,2%)

É.T. = Écart-Type

Causal Presuppositions

- *Interference*: 4th down decision in one game does not affect winning in another game
- *Consistency*: All possibilities of 4th downs are accounted (Intervention well defined)
- *Positivity*: Every game chosen had a 4th down and empirical positivity holds
- *Conditional Exchangeability*: Adjusted for potential confounders in TMLE and in propensity score model

Propensity Score Models

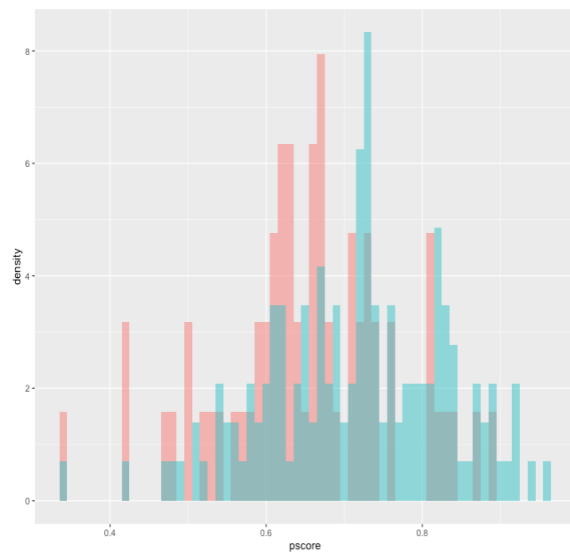
- *Continuous variables*: Spread Line, Temperature (F), Wind (mph),
- *Categorical Variables*: Roof, Surface, Opposing Team Defensive Team Rating, Offensive Team Rating and Field Location
- Standardized all continuous variables
- Modelled continuous variables with exposure to assess Linearity

Propensity Score Models

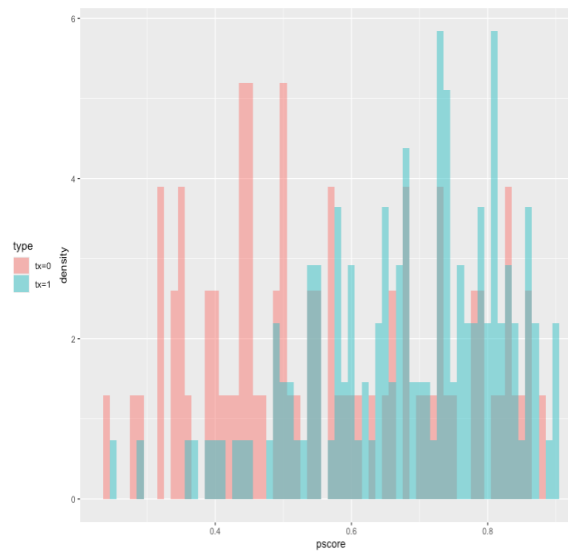
- Logistic regression model with categorical/continuous variables as predictors and exposure as outcome
- *IND*: Linear term for wind, quadratic term for spread line and cubic term for temperature
- *JAX*: Quartic term for wind, linear term for spread line and quadratic term for temperature
- *KC*: Quadratic term for wind, linear term for spread line and linear term for temperature

Matching

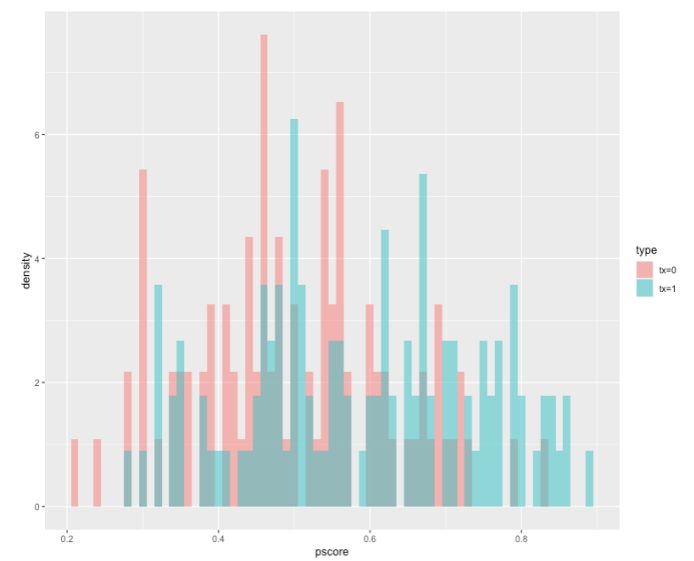
No extreme π (<0.05 or >0.95) for all three teams



IND



JAX



KC

Matching

- For every team, one-to-one matching was used with replacement
- *IND*: Caliper = 0,09, Proportion of Treated observations lost = 3/137, Kolmogorov Smirnov min $p = 0,088$
- *JAX*: Caliper = 0,1, Proportion of Treated observations lost = 12/144, Kolmogorov Smirnov min $p = 0,077$
- *KC*: Caliper = 0,08, Proportion of Treated observations lost = 14/112, Kolmogorov Smirnov min $p = 0,102$



IPTW

- For each team, I modelled a generalized linear model with outcome, exposure as predictor and corresponding weights
- Truncation with Gruber bound
- Truncation with Percentile (at 0.05% and 0.95%)
- Truncation leads to same conclusions and similar estimates

Table 1 after IPTW for IND

<i>Game characteristics</i>		<i>Exposure = 0</i>	<i>Exposure = 1</i>	<i>Standardized Mean Difference</i>
Temperature (F) ; Moyenne (É.T.)		57,12 (15,72)	56,82 (17,55)	0,018
Spread line ; Moyenne (É.T.)		-0,36 (5,93)	-0,59 (5,97)	0,038
Wind (MPH) ; Moyenne (É.T.)		7,75 (3,90)	7,86 (4,69)	0,025
Outcome ; N (%)				0,720
	Loss	59,9 (27,0%)	128, 6 (60,7%)	
	Win	161,6 (73,0%)	83,4 (39,3%)	
Field Location ; N (%)				0,061
	Away	113,5 (51,2%)	102,2 (48,2%)	
	Home	108,0 (48,8%)	109,8 (51,8%)	
Surface ; N (%)				0,079
	Turf	140,1 (63,2%)	142,1 (67,0%)	
	Grass	81,4 (36,8%)	69,9 (33,0%)	
Roof ; N (%)				0,039
	Closed/Dome	103,6 (46,8%)	103,2 (48,7%)	
	Outdoor/Open	117,9 (53,2%)	108,8 (51,3%)	
Opposing team defensive team rating ; N (%)				0,032
	Top 16	112,0 (50,6%)	110,6 (52,2%)	
	Bottom 16	109,4 (49,4%)	101,4 (47,8%)	
Offensive team rating ; N (%)				0,047
	Top 16	150,2 (67,8%)	139,0 (65,5%)	
	Bottom 16	71,3 (32,2%)	73,0 (34,4%)	

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Table 1 after IPTW for JAX

<i>Game characteristics</i>		<i>Exposure = 0</i>	<i>Exposure = 1</i>	<i>Standardized Mean Difference</i>
Temperature (F) ; Moyenne (É.T.)		66,64 (14,40)	65,07 (15,00)	0,107
Spread line ; Moyenne (É.T.)		-3,04 (5,00)	-3,47 (6,25)	0,076
Wind (MPH) ; Moyenne (É.T.)		8,04 (4,14)	8,11 (4,43)	0,017
Outcome ; N (%)				0,744
	Loss	89,2 (43,6%)	161,1 (77,7%)	
	Win	115,4 (56,4%)	46,3 (22,3%)	
Field Location ; N (%)				0,003
	Away	101,1 (49,4%)	102,2 (49,3%)	
	Home	103,5 (50,6%)	105,3 (50,7%)	
Surface ; N (%)				0,002
	Turf	44,8 (21,9%)	45,3 (21,8%)	
	Grass	159,7 (78,1%)	162,2 (78,2%)	
Roof ; N (%)				0,063
	Closed/Dome	24,8 (12,1%)	29,6 (14,3%)	
	Outdoor/Open	179,7 (87,9%)	177,8 (85,7%)	
Opposing team defensive team rating ; N (%)				0,005
	Top 16	101,9 (49,8%)	102,8 (49,5%)	
	Bottom 16	102,7 (50,2%)	104,7 (50,5%)	
Offensive team rating ; N (%)				0,026
	Top 16	60,0 (29,3%)	58,4 (28,1%)	
	Bottom 16	144,5 (70,7%)	149,1 (71,9%)	

É.T. = Écart-Type

Table 1 after IPTW for KC

<i>Game characteristics</i>		<i>Exposure = 0</i>	<i>Exposure = 1</i>	<i>Standardized Mean Difference</i>
Temperature (F) ; Moyenne (É.T.)		56,84 (18,02)	56,50 (18,05)	0,018
Spread line ; Moyenne (É.T.)		3,40 (6,17)	3,21 (7,15)	0,028
Wind (MPH) ; Moyenne (É.T.)		7,88 (4,46)	8,17 (4,65)	0,064
Outcome ; N (%)				0,324
	Loss	45,7 (22,8%)	76,9 (37,5%)	
	Win	154,8 (77,2%)	128,4 (62,5%)	
Field Location ; N (%)				0,011
	Away	98,1 (48,9%)	99,3 (48,4%)	
	Home	102,3 (51,1%)	106,0 (51,6%)	
Surface ; N (%)				0,009
	Turf	46,0 (23,0%)	47,9 (23,4%)	
	Grass	154,4 (77,0%)	157,4 (76,6%)	
Roof ; N (%)				0,024
	Closed/Dome	18,1 (9,1%)	17,2 (8,4%)	
	Outdoor/Open	182,2 (90,9%)	188,1 (91,6%)	
Opposing team defensive team rating ; N (%)				0,010
	Top 16	87,9 (43,8%)	89,0 (43,4%)	
	Bottom 16	112,5 (56,2%)	116,3 (56,6%)	
Offensive team rating ; N (%)				0,009
	Top 16	171,7 (85,7%)	175,2 (85,3%)	
	Bottom 16	28,7 (14,3%)	30,1 (14,7%)	

É.T. = Écart-Type



TMLE with SL

- Same covariables used in TMLE as in Propensity Score Model
- *Fewer learners*: Effective sample size* for all three teams less than 500
- *Less flexible learners*: Effective sample size less than 500 (SL.GLM and SL.mean)
- *Learners that do not force linear relationship*: Continuous covariates (SL.earth)

* (Phillips & van der Laan & Lee & Gruber, 2023)



Causal Estimations for MRD

Method	<i>Estimation</i> <i>[CI 95%]</i>	Method	<i>Estimation</i> <i>[CI 95%]</i>	Method	<i>Estimation</i> <i>[CI 95%]</i>
TMLE SL	-0,300 [-0,433, -0,184]	TMLE SL	-0.299 [-0,456, -0,165]	TMLE SL	-0,151 [-0,253, -0,035]
IPTW	-0,336 [-0,412, -0,066]	IPTW	-0,341 [-0,583, -0,183]	IPTW	-0,147 [-0,379, -0,052]

IND

JAX

KC

Causal interpretation for MRD

- If the causal assumptions are respected, we conclude (for my three teams):
- The estimated MRD for 4th down attempts is negative. This means that, on average, if the team had attempted a 4th down play in a game between the 2011-2024 seasons, **their probability of winning would have decrease** compared to if they had not attempted one
- The confidence interval excludes zero, suggesting that 4th down attempts were ineffective in increasing winning on average across all games



Causal Estimations for ATT

Method	<i>Estimation</i> <i>[CI 95%]</i>	Method	<i>Estimation</i> <i>[CI 95%]</i>	Method	<i>Estimation</i> <i>[CI 95%]</i>
TMLE SL	-0,294 [-0,477, -0,174]	TMLE SL	-0,316 [-0,564, -0,149]	TMLE SL	-0,177 [-0,307, -0,027]
Match	-0,373 [-0,552, -0,194]	Match	-0,318 [-0,476, -0,160]	Match	-0,189 [-0,321, -0,057]

IND

JAX

KC

Causal interpretation for ATT

- If the causal assumptions are respected, we conclude (for my three teams):
- The estimated ATT for 4th down attempts is negative. This means that, on average, in games where a 4th down play was attempted between the 2011-2024 seasons, the probability of winning would have been higher if they had not attempted it
- The confidence interval excludes zero, suggesting that 4th down attempts were ineffective in increasing the probability of winning in games where they were used
- Adopting a more conservative 4th down strategy in those games could have led to a higher chance of winning



Causal Estimations for ATC

Method	<i>Estimation</i> <i>[CI 95%]</i>	Method	<i>Estimation</i> <i>[CI 95%]</i>	Method	<i>Estimation</i> <i>[CI 95%]</i>
TMLE SL	-0,306 [-0,447,-0,096]	TMLE SL	-0,295 [-0,474, -0,137]	TMLE SL	-0,129 [-0,283, -0,005]
IND		JAX		KC	

Causal interpretation for ATC

- If the causal assumptions are respected, we conclude (for my three teams):
- The estimated ATC for 4th down attempts is negative. This means that, on average, in games where a 4th down play was not attempted between the 2011-2024 seasons, the probability of winning would have decreased if they had attempted one
- The confidence interval excludes zero, indicating that 4th down attempts would have been ineffective in increasing the probability of winning in games where they were not used
- Adopting a more aggressive 4th down strategy in those games could have led to a lower chance of winning

Does attempting 4th down plays in an NFL game cause more winning?

- Decreased probability of winning
- Lower estimates for best team (KC)
- National Football League

References

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